

Data Flow Diagram

Date	8 july 2026
Team ID	
Project Name	StudyAI-Smart AI educate comapanion&Quiz Generator
Maximum Marks	2 Marks

Data Flow Diagram

This Data Flow Diagram shows how a student's data moves through StudyAI — starting from login and file upload, through the Flask backend and Groq AI engine that generate summaries, flashcards, and quizzes, and finally into Firebase Firestore where user data, content, and quiz scores are stored and later retrieved for the analytics dashboard.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person or system outside StudyAI that sends or receives data e.g., the Student (User) and the Groq API.
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data e.g., 1 Authentication, 2 AI Generation, 3 Analytics.
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use e.g., Firebase Firestore (users, documents, quiz scores).
Labeled arrow	Data Flow	The movement of data between entities, processes, and stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

StudyAI — Data Flow Diagram (Level 1)

